

# E-09-bypass-no-frame v1: can $W$ , enstrophy, dissipation and $L^3$ separate $W_-$ without spectral frame?

ZFL Lab

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MISSION: Determine if  $W, \|\omega\|_2, \|\nabla\omega\|_2, \|u\|_3$  contain enough information to separate  $W_+$  from  $W_-$  without diagonalizing  $S$ . Attack  $W_-$  directly.

INPUT:  $\mathcal{C}_{PROVEN} = \{C_E^{weak} : \limsup \int_0^t W = +\infty, C_{L^3} : \limsup \|u\|_3 = +\infty\}$ . Identities:

$$W = \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla\omega\|_2^2 \text{ EXACT}$$

$$|W| \leq C_S \|u\|_3 \|\nabla\omega\|_2^2$$

with  $W = W_+ - W_-$ ,  $W_{\pm} \geq 0$ ,  $\int_0^{T^*} W_+ = \infty$  under  $C_E^{weak}$  if  $W_- \in L^1$  not proven.

FORBIDDEN:  $e_i, \lambda_i, \eta_i, \eta_-, D_t e_i$  any spectral frame.  $\|u\|_3 \rightarrow \infty \Rightarrow W_+ \rightarrow \infty$ .  $\|u\|_3 \rightarrow \infty \Rightarrow W_-$  controlled.  $|W| \leq C_S \|u\|_3 \|\nabla\omega\|_2^2 \Rightarrow W_- \in L_t^1$ .  $C_E^{weak} \Rightarrow C_E^{strong}$  without new estimate.

AUDIT:

1. What  $|W| \leq C_S \|u\|_3 \|\nabla\omega\|_2^2$  actually gives:

Since  $W_- \leq |W|$ ,

$$W_-(t) \leq C_S \|u(t)\|_3 \|\nabla\omega(t)\|_2^2.$$

Integrate:

$$\int_0^{T^*} W_- \leq C_S \int_0^{T^*} \|u\|_3 \|\nabla\omega\|_2^2.$$

Under  $C_{L^3}$ ,  $\|u\|_3$  may diverge, so RHS may diverge too. No  $L^1$  control. Trampa blindada:

$$|W| \leq C_S \|u\|_3 \|\nabla\omega\|_2^2 \not\Rightarrow W_- \in L_t^1.$$

2. Enstrophy identity vs sign:

From  $W = \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla\omega\|_2^2$ ,

$$\int_0^t W = \frac{1}{2} \|\omega(t)\|_2^2 - \frac{1}{2} \|\omega(0)\|_2^2 + \nu \int_0^t \|\nabla\omega\|_2^2.$$

Thus  $C_E^{weak}$  i.e.  $\int_0^t W \rightarrow \infty$  implies

$$\frac{1}{2} \|\omega(t)\|_2^2 + \nu \int_0^t \|\nabla\omega\|_2^2 \rightarrow \infty.$$

But this uses total  $W$ , not  $W_{\pm}$ . Since  $W = W_+ - W_-$ ,

$$\int W_+ = \int W + \int W_- = \left( \frac{1}{2} \|\omega\|_2^2 + \nu \int \|\nabla\omega\|_2^2 \right) + \int W_- - \frac{1}{2} \|\omega(0)\|_2^2.$$

Hence  $\int W_+ = \infty$  can be due to  $\int W_- = \infty$  plus even larger growth of enstrophy, not forbidden.

Goal would be inequality

$$\boxed{W_- \leq F(\|u\|_3, \|\omega\|_2, \|\nabla\omega\|_2)}$$

with  $\int_0^{T_*} F < \infty$  under  $C_E^{weak} \wedge C_{L^3}$ , or at least  $F$  small enough that  $W_+$  divergence cannot be cancelled.

Search for such  $F$  without frame:

Direct expression  $W = \int \omega^T S \omega = \int \omega \cdot (\omega \cdot \nabla u)$ ? Using  $S$  symmetric part of  $\nabla u$ , but  $\omega^T \Omega \omega = 0$ , so  $W = \int \omega^T \nabla u \omega$ . Integration by parts:  $W = \int \omega_i \omega_j \partial_j u_i$ . No sign information.

No pointwise bound for negative part from integration by parts alone because  $S$  indefinite with  $\text{tr } S = 0$  not used if we avoid eigenvalues.

Consider splitting  $u = P_{\leq N} u + P_{> N} u$  Littlewood-Paley:  $\|u\|_3$  controls low frequencies, but high frequencies contribute to  $\|\nabla\omega\|_2$ . Still no sign.

Potential candidate  $F$  using interpolation:  $\|\omega\|_2^2 \leq \|u\|_3 \|\nabla\omega\|_{3/2}$ ? By Sobolev? Actually  $\|\omega\|_2 \leq C \|u\|_3^a \|\nabla\omega\|_2^{1-a}$ ? Need check.

But to get  $W_-$  control, need lower bound on  $W$ , not upper:  $W \geq -C_S \|u\|_3 \|\nabla\omega\|_2^2$ , which gives  $W_- \leq C_S \|u\|_3 \|\nabla\omega\|_2^2$  again, same trap.

Thus without extra quantity,  $F$  is exactly that.

3. Is information sufficient?

We have only: -  $W$  exact decomposition  $W = \dot{E} + \nu D$ ,  $E = \frac{1}{2} \|\omega\|_2^2$ ,  $D = \|\nabla\omega\|_2^2 - |W| \leq C_S \|u\|_3 D$

From these, can we deduce control of  $W_-$ ? No. System of inequalities:

$W_+ - W_- = \dot{E} + \nu D$   $W_+ + W_- = |W| + 2 \min(W_+, W_-)$  Not determined. Actually  $|W| = |W_+ - W_-|$ , so  $W_+ + W_-$  not controlled.

We have 2 equations for 3 unknowns  $W_+, W_-, E, D, \|u\|_3$  loosely coupled. No sign separation.

Thus conjecture:

$$\boxed{C_E^{weak} \wedge C_{L^3} \text{ with only } W, \|\omega\|_2, \|\nabla\omega\|_2, \|u\|_3 \text{ insufficient to bound } W_- \text{ in } L_t^1.}$$

Need new quantity  $C_X$ .

Possible  $C_X$  candidates bypassing frame: - pressure:  $p$  structure,  $\int p |\omega|^2$ ? - helicity  $H = \int u \cdot \omega$ ? - higher  $L^p$  for  $\omega$ ? - vorticity direction regularity via  $\nabla \xi$  but without  $e_i$ ?

But mission of E-09 is to test, not to propose  $C_X$  yet.

OUTPUT:

$$\boxed{W = \frac{1}{2} \frac{d}{dt} \|\omega\|_2^2 + \nu \|\nabla\omega\|_2^2 \text{ and } |W| \leq C_S \|u\|_3 \|\nabla\omega\|_2^2 \text{ EXACT, but give only } W_- \leq C_S \|u\|_3 \|\nabla\omega\|_2^2}$$

$$\boxed{\int_0^{T_*} C_S \|u\|_3 \|\nabla\omega\|_2^2 \text{ may diverge under } C_{L^3}, \text{ so no } W_- \in L_t^1}$$

$$\boxed{\text{No inequality of form } W_- \leq F(\|u\|_3, \|\omega\|_2, \|\nabla\omega\|_2) \text{ with } F \in L_t^1 \text{ found from identities alone}}$$

$$\boxed{C_E^{weak} \wedge C_{L^3} \stackrel{?}{\Rightarrow} \text{control } W_- = \text{OPEN, likely insufficient}}$$

Classification: PARTIAL/OPEN — identities isolated, insufficiency identified, need  $C_X$  beyond current 4 quantities.

$C_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$  intact. E-05 to E-08 frozen. No spectral objects used.

Next: define search for  $C_X$  — pressure-vorticity coupling or helicity or direction regularity without eigenframe.